EC–1848

B. Tech. (First Semester)
EXAMINATION, 2025-26
ELECTRONICS ENGG.

Time : Two Hours

Maximum Marks : 100

Note : Attempt questions from all Sections as directed.

Section—A
(Very Short Answer Type Questions)

Note : Attempt any *seven* questions. Each question carries 8 marks. $7 \times 8 = 56$

1. Explain about Diode Capacitance and Junction Capacitance.
2. Explain the diode current equation. A GE diode at T = 300 K conducts 5 mA at V = 0.35 V. Determine the diode current at V = 0.4 V.

P. T. O.

3. Explain the working of Full Wave Bridge Rectifier Circuit.

4. Draw input and output characteristics of common collector configuration of BJT.

5. Explain the various parameters of operation amplifiers.

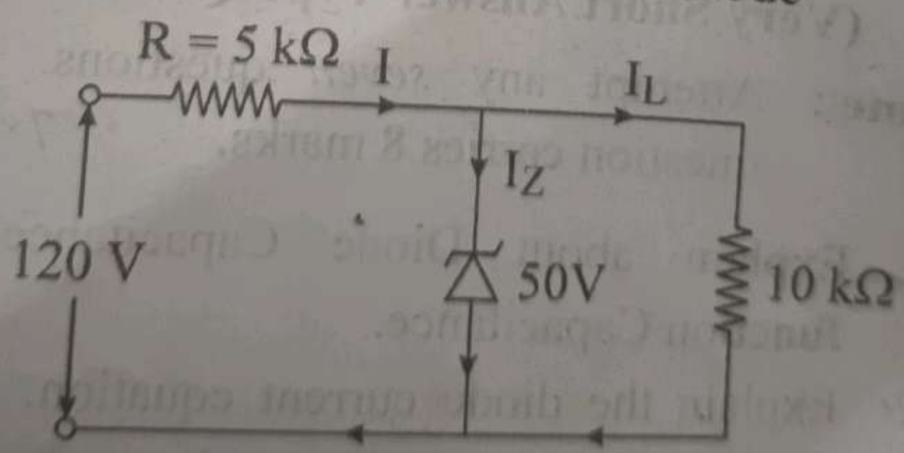
6. Explain the various types of Biasing circuits.

7. For the circuit given below, find :

   (A) Output voltage

   (B) The voltage drop across series resistance

   (C) The current through Zener diode



8. Discuss the working of C.R.O. with block diagram.

9. Derive the relationship between α and β.

10. Define Inverting and Non-Inverting Operation Amplifier.

Section—B

(Short Answer Type Questions)

Note : Attempt any *two* questions. Each question carries 12 marks. 2×12=24

1. Explain the working principle of OP-Amp as Integrator and Differentiator.

2. Explain the working with suitable diagram of Negative Shunt Clipper and Positive Clamper Circuit.

3. (a) Minimize the following function using K-Map :

$$F(A, B, C, D) = \Sigma m\,(1, 3, 4, 6, 8, 9, 11, 13, 15) + \Sigma d\,(0, 2, 14)$$

(b) Minimize the following function using Boolean Algebra :

$$F(X, Y, Z) = X'Y + YZ' + YZ + XY'Z'$$

4. Explain the working of digital multimeter with a block diagram.

Section—C

(Long Answer Type Questions)

Note : Attempt any *one* question. Each question carries 20 marks. $1 \times 20 = 20$

1. Draw and explain the operation of P-channel enhancement type MOSFET.

2. (a) Explain OP Amp as unity gain configuration.

   (b) Explain half wave rectifier circuit.

3. Discuss the operation of P-N diode in forward bias and reverse bias and also discuss V-I characteristics with knee voltage and breakdown voltage.

4. Convert the following :

   (a) $(12025.056)_{10} = (\ )_2$

   (b) $(345.706)_8 = (\ )_{10}$

   (c) $(3A4.BC)_{16} = (\ )_8$

   (d) $(10110011011)_2 = (\ )_8$

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